

SDS-PAGE Gel Preparation

Laboratory Report — Resolving and Stacking Gel Casting

Analyst: Gontse Bridgett Baloyi
Technique category: Protein Biochemistry / Method Preparation

1. Objective

To calculate and prepare resolving and stacking polyacrylamide gels of defined percentage and volume for subsequent SDS-PAGE protein separation.

2. Introduction

Before SDS-PAGE can separate proteins, a polyacrylamide gel must be cast consisting of a resolving gel (typically 8–15% acrylamide, chosen based on target protein size) topped with a low-percentage stacking gel that concentrates samples into a sharp starting band. Reagent volumes must be calculated precisely, and ammonium persulfate (APS) and TEMED — which together initiate and catalyse polymerisation — are always added last, immediately before pouring, since polymerisation begins within minutes.

3. Materials

- 30% acrylamide/bis-acrylamide stock solution (29:1)
- Resolving gel buffer: 1.5M Tris-HCl, pH 8.8
- Stacking gel buffer: 1.0M Tris-HCl, pH 6.8
- 10% SDS, 10% APS, TEMED
- Distilled water

4. Method

- Assembled glass plates in the casting stand and checked for leaks.
- Combined resolving gel components (acrylamide, buffer, SDS, water) at the target percentage.
- Added APS and TEMED last; mixed gently and poured immediately.
- Overlaid with water to exclude air and create a level gel front; allowed to polymerise (~30–45 min).
- Removed the overlay, prepared the stacking gel (fixed at ~4%), and poured on top with a comb inserted.
- Allowed the stacking gel to polymerise (~20–30 min) before use.

5. Results

Reagent volumes were calculated for a 10% resolving gel (10 mL total volume) and a standard 4% stacking gel (5 mL total volume), and were also compared across the common resolving gel percentage range (8–15%).

Component	Resolving (10%, 10 mL)	Stacking (4%, 5 mL)
30% Acrylamide/bis	3.33 mL	0.67 mL

Buffer	2.50 mL	1.25 mL
10% SDS	0.10 mL	0.05 mL
Water	3.96 mL	2.98 mL
10% APS	0.10 mL	0.05 mL
TEMED	0.004 mL	0.003 mL

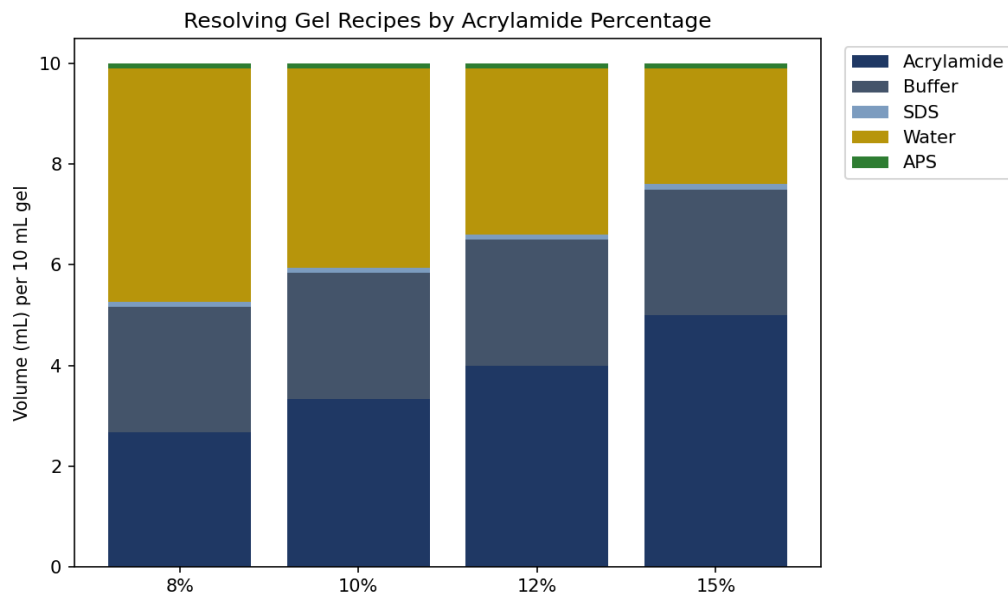


Figure 1. Resolving gel component volumes across common acrylamide percentages (8–15%), per 10 mL gel.

6. Discussion

As resolving gel percentage increases, the proportion of acrylamide stock required rises linearly while the water volume decreases correspondingly, since buffer, SDS, and APS remain fixed proportions of total gel volume regardless of percentage. The slightly lower TEMED volume used at higher percentages reflects faster polymerisation kinetics in denser gels, reducing the amount of catalyst needed to achieve a workable polymerisation time. Selecting the correct resolving gel percentage for a given protein's expected molecular weight range is important for achieving good separation — lower percentages resolve larger proteins better, while higher percentages resolve smaller proteins better.

7. Conclusion

This exercise demonstrated systematic calculation of SDS-PAGE gel recipes across a range of acrylamide percentages, reinforcing the relationship between gel percentage, reagent proportions, and the resulting protein separation range achievable on a cast gel.